## Model

| Layer | Shape | Elements | Bytes (FP32) |
|---|---|---|---|
| input | $B \times 3 \times S \times S$ | $3BS^2$ | $12\,BS^2$ |
| Conv 7×7 s2 + ReLU | $B \times 32 \times S/2 \times S/2$ | $8BS^2$ | $32\,BS^2$ |
| MaxPool 3×3 s2 p1 | $B \times 32 \times S/4 \times S/4$ | $2BS^2$ | $8\,BS^2$ |
| Conv 5×5 + ReLU | $B \times 64 \times S/4 \times S/4$ | $4BS^2$ | $16\,BS^2$ |
| Conv 3×3 s2 + ReLU | $B \times 128 \times S/8 \times S/8$ | $2BS^2$ | $8\,BS^2$ |
| Conv 1×1 + ReLU | $B \times 256 \times S/8 \times S/8$ | $4BS^2$ | $16\,BS^2$ |
| Conv 3×3 s2 + ReLU | $B \times 256 \times S/16 \times S/16$ | $BS^2$ | $4BS^2$ |
| Conv 1×1 + ReLU | $B \times 512 \times S/16 \times S/16$ | $2BS^2$ | $8\,BS^2$ |
| GAP | $B \times 512$ | $512B$ | $2048B$ |
| Linear + ReLU | $B \times 256$ | $256B$ | $1024B$ |
| Linear | $B \times 100$ | $100B$ | $400\,B$ |

## FLOPs

1 MAC = 2 FLOPs

Conv: $2 \cdot B \cdot C_{out} \cdot H_{out} \cdot W_{out} \cdot C_{in} \cdot k^2$

Linear: $2 \cdot B \cdot F_{in} \cdot F_{out}$

ReLU: 1 FLOP per element $BCHW$ or $B \cdot F$

MaxPool: $(k^2-1)$ compares per window $B\,C_{out}\,H_{out}\,W_{out}\,(k^2-1)$

GAP: $(HW-1)$ add + 1 division per channel $BCHW$

FLOPs

| layer | FLOPs | Term |
|---|---|---|
| Conv 7x7 s2 | $2B \cdot 32 \cdot S/2 \cdot S/2 \cdot 3 \cdot 7^2$ | $2352\ BS^2$ |
| ReLU | $B \cdot 32 \cdot S/2 \cdot S/2$ | $8\ BS^2$ |
| MaxPool 3x3 | $B32\ S/4 \cdot S/4 \cdot (3^2 - 1)$ | $16\ BS^2$ |
| Conv 5x5 | $2B \cdot 64 \cdot S/4 \cdot S/4 \cdot 32 \cdot 5^2$ | $6400\ BS^2$ |
| ReLU | $B \cdot 64 \cdot S/4 \cdot S/4$ | $4\ BS^2$ |
| Conv 3x3 s2 | $2B \cdot 128 \cdot S/8 \cdot S/8 \cdot 64 \cdot 3^2$ | $2304\ BS^2$ |
| ReLU | $B \cdot 128 \cdot S/8 \cdot S/8$ | $2\ BS^2$ |
| Conv 1x1 | $2B \cdot 256 \cdot S/8 \cdot S/8\ 128 \cdot 1^2$ | $1024\ BS^2$ |
| ReLU | $B \cdot 256 \cdot S/8 \cdot S/8$ | $4\ BS^2$ |
| Conv 3x3 s2 | $2B\ 256 \cdot S/16 \cdot S/16 \cdot 256 \cdot 3^2$ | $4608\ BS^2$ |
| ReLU | $B \cdot 256 \cdot S/16 \cdot S/16$ | $1\ BS^2$ |
| Conv 1x1 | $2B \cdot 512 \cdot S/16 \cdot S/16 \cdot 256 \cdot 1^2$ | $1024\ BS^2$ |
| ReLU | $B \cdot 512 \cdot S/16 \cdot S/16$ | $2\ BS^2$ |
| GAP | $B \cdot 512 \cdot S/16 \cdot S/16$ | $2\ BS^2$ |
| Linear | $2B \cdot 512 \cdot 256$ | $262144\ B$ |
| ReLU | $B \cdot 256$ | $256\ B$ |
| Linear | $2B \cdot 256 \cdot 100$ | $51200\ B$ |
| TOTAL: | | $17751\ BS^2 + 313600\ B$ |

# Peak memory

## model weights

| Layer | Params |
|---|---|
| Conv 7×7 | $32 \cdot 3 \cdot 7 \cdot 7 = 4704$ |
| Conv 5×5 | $64 \cdot 32 \cdot 5 \cdot 5 = 51200$ |
| Conv 3×3 | $128 \cdot 64 \cdot 3 \cdot 3 = 73728$ |
| Conv 1×1 | $256 \cdot 128 = 32768$ |
| Conv 3×3 | $256 \cdot 256 \cdot 3 \cdot 3 = 589824$ |
| Conv 1×1 | $512 \cdot 256 = 131072$ |
| Linear 512→256 + bias | $512 \cdot 256 + 256 = 131328$ |
| Linear 256→100 + bias | $256 \cdot 100 + 100 = 25700$ |
| TOTAL: | 1040.324 |

$W = 4 \cdot P = 4161296$ B

Peak at MaxPool : At each step we store
- input to layer
- output of layer
- input X of model
+ if layer = MaxPool
  - tmp indices for maxpool

| Buffer | Bytes |
|---|---|
| input | $12BS^2$ |
| output conv 7×7 | $32BS^2$ |
| output pool | $8BS^2$ |
| indices pool, int64 | $2BS^2 \cdot 8$ Bytes $= 16BS^2$ |
| TOTAL | $68BS^2$ |

Peak memory : $4.161.296 + 68BS^2$

## Bytes moved

Each layer: read input + write output + model weights once

| Layer | in+out, elements | bytes |
|---|---|---|
| Conv 7x7 | $(3+8)BS^2$ | $44\,BS^2$ |
| MaxPool, maps | $(8+2)BS^2$ | $40\,BS^2$ |
| MaxPool, indices | $2\,BS^2$ | $8\,BS^2$ |
| ReLU | $(8+8)BS^2$ | $64\,BS^2$ |
| Conv 5x5 | $(2+4)BS^2$ | $24\,BS^2$ |
| ReLU | $(4+4)BS^2$ | $32\,BS^2$ |
| Conv 3x3 | $(4+2)BS^2$ | $24\,BS^2$ |
| ReLU | $(2+2)BS^2$ | $16\,BS^2$ |
| Conv 1x1 | $(2+4)BS^2$ | $24\,BS^2$ |
| ReLU | $(4+4)BS^2$ | $32\,BS^2$ |
| Conv 3x3 | $(4+1)BS^2$ | $20\,BS^2$ |
| ReLU | $(1+1)BS^2$ | $8\,BS^2$ |
| Conv 1x1 | $(1+2)BS^2$ | $12\,BS^2$ |
| ReLU | $(2+2)BS^2$ | $20\,BS^2$ |
| GAP | $2BS^2+512B$ | $8BS^2+2048B$ |
| Linear | $(512+256)B$ | $3072\,B$ |
| ReLU | $(256+256)B$ | $2048\,B$ |
| Linear | $(256+100)B$ | $1424\,B$ |
| TOTAL: | | $380\,BS^2+8592B+$ $+4161296$ bytes |

ReLU does not allocate new map, but still reads and writes same map

## Latency

$$T = T_0 + \max\left(F/\pi, Q/\beta\right)$$

- $T_0$: launch overhead
- $F$: required FLOPs
- $\pi$: achieved FLOPs/s
- $Q$: moved memory bytes
- $\beta$: measured bandwidth bytes/s

$$\theta = \{T_0, \pi, \beta\}$$

## Energy

$$E = P_0 T + e_Q \cdot Q$$

- $P_0$: power for screen and side ops
- $T$: latency
- $e_Q$: measured power per moved byte
- $Q$: moved bytes.

$$\theta = \{P_0, e_Q\}$$